

EDUCATION

Ph.D. in Physics, Harvard University
M.A. in Physics, Harvard University
B.A. in Physics, Cornell University

Ph.D. Thesis: *Hidden Dynamics of Static Friction* 2020
 Ph.D. Advisor: Shmuel M Rubinstein 2016
 Undergraduate Research Advisor: Itai Cohen 2012

RESEARCH EXPERIENCE

ARIA R&D Creator (Independent Researcher) 2025–2027
Topics: Learning Metamaterials | Distributed & Analog Machine Learning

Postdoctoral Fellow, University of Pennsylvania with Douglas J Durian & Andrea J Liu 2020–2025
Topics: Emergent Physical Learning | Granular Flows | ML in Experimental Science

Visiting Researcher, EPFL, Switzerland with John M Kolinski 2018
Topic: Developed ultrafast ($\geq$ MHz) imaging technique for any camera

FUNDING AND AWARDS

Grants & Fellowships

ARIA Opportunity Seed Grant £500k (~\$650k) 2025–2027
 AI Fellow, U Pennsylvania \$8k/year, 2025–2026
 Center for Soft and Living Matter Fellow, U Pennsylvania \$5k/year, 2025
 Data Science Postdoctoral Fellow, U Pennsylvania \$5k/year, 2022–2024
 Smith Family Fellowship, Harvard U \$90k, 2015–16
 Purcell Fellowship, Harvard U \$90k, 2014–15

Research & Teaching Recognition

1st place GSNP Postdoctoral Presentation Award APS March Meeting 2024
 1st place Meeting-Wide Postdoctoral Poster Competition APS March Meeting 2023
 Herbert B. Callen Memorial Prize University of Pennsylvania 2023
 2nd place, title: [Learning Networks on the Radio](#) MRSEC National Science Slam 2022
 Editors' Suggestion, Physical Review Applied Dillavou et al. 2022
 Rising Star in Soft and Biological Matter University of Chicago 2021
 Editors' Suggestion, Physical Review Letters Dillavou & Rubinstein 2018
 Bok Center Certificate of Teaching Excellence Harvard University Spring 2018

PUBLICATIONS

▪ equal contribution • work performed as an undergraduate

Lead Author

- [1] **S Dillavou**, JW Rocks, JF Wycoff▪, AJ Liu, DJ Durian. *Double descent without digital computation* PNAS, 2026
- [2] **S Dillavou**, S Mishra. *Physical Reinforcement Learning* RL in Big Worlds Workshop at RLC, 2026
- [3] **S Dillavou**▪, M Guzman▪, AJ Liu, DJ Durian. *Understanding and Embracing Imperfection in Physical Learning Networks* arXiv 2505.22887
- [4] JM Hanlan▪, **S Dillavou**▪, AJ Liu, DJ Durian. *Cornerstones are the Key Stones: Using Interpretable Machine Learning to Probe the Clogging Process in 2D Granular Hoppers* Soft Matter, 2025
- [5] D Hathcock▪, **S Dillavou**▪, JM Hanlan, DJ Durian, Y Tu. *Stochastic Dynamics of Granular Hopper Flows: A Configurational Mode Controls the Stability of Clogs* Physical Review E, 2025
- [6] (Perspective) **S Dillavou**. *Harnessing Machine Learning to Guide Scientific Understanding* Physics Magazine, 2024
- [7] **S Dillavou**, BD Beyer•, M Stern, AJ Liu, MZ Miskin▪, DJ Durian▪. *Machine learning without a processor: Emergent learning in a nonlinear analog network* PNAS, 2024
- [8] **S Dillavou**, JM Hanlan, AT Chieco, H Xiao, S Fulco, KT Turner, DJ Durian. *Bellybutton: accessible and customizable deep-learning image segmentation* Scientific Reports, 2024
- [9] **S Dillavou**, BD Beyer•, M Stern, MZ Miskin, AJ Liu, DJ Durian. *Nonlinear Classification Without a Processor* NeurIPS ML with New Compute Paradigms Workshop, 2023
- [10] **S Dillavou**, BD Beyer•, M Stern, MZ Miskin, AJ Liu, DJ Durian. *Circuits That Train Themselves: Decentralized, Physics-Driven Learning* Proceedings SPIE, AI & Optical Data Sciences IV, 2023
- [11] **S Dillavou**, Y Bar-Sinai, MP Brenner, SM Rubinstein. *Beyond Quality and Quantity: Spatial Distribution of Contact Encodes Frictional Strength* Physical Review E, 2022
- [12] **S Dillavou**, M Stern, AJ Liu, DJ Durian. *Demonstration of Decentralized Physics-Driven Learning* Editors' Suggestion, Physical Review Applied, 2022
- [13] **S Dillavou** and SM Rubinstein. *Shear Controls Frictional Aging by Erasing Memory* Physical Review Letters, 2020
- [14] **S Dillavou**, SM Rubinstein, JM Kolinski. *The Virtual Frame Technique: Ultrafast Imaging with Any Camera* Optics Express, 2019

- [15] **S Dillavou** and SM Rubinstein. *Nonmonotonic Aging and Memory in a Frictional Interface* [Editors' Suggestion, Physical Review Letters](#), 2018

Additional

- [16] KA Murphy, **S Dillavou**, DS Bassett. *Comparing the Information Content of Probabilistic Representation Spaces* [Transactions on ML Research](#), 2025
- [17] AJ Gerra[✉], CC Jones[✉], **S Dillavou**, JM Hanlan, J Radzio, PE Arratia, DJ Durian. *Equation of motion for taut-line buzzers* [Physical Review Applied](#), 2024
- [18] T Martin, **S Dillavou**. *Calculations without Math: "Smart Instruments" and the Transposition of Complex Shapes in the Wooden Boat Workshop* [J of Cultural Cognitive Science](#), 2024
- [19] M Stern, **S Dillavou**, D Jayaraman, DJ Durian, AJ Liu. *Training Self-Learning Circuits for Power-Efficient Solutions* [APL Machine Learning](#), 2024
- [20] M Stern, **S Dillavou**, D Jayaraman, DJ Durian, AJ Liu. *Contrastive Power-Efficient Physical Learning in Resistor Networks* [NeurIPS ML with New Compute Paradigms Workshop](#), 2023
- [21] W Steinhardt, **S Dillavou**, M Agajanian[✉], SM Rubinstein, EE Brodsky. *Seismological Stress Drops for Confined Ruptures Are Invariant to Normal Stress* [Geophysical Research Letters](#), 2023
- [22] A Srivastava ... **S Dillavou** ... Z Wu (100s of authors). *Beyond the Imitation Game: Quantifying and Extrapolating the Capabilities of Language Models* [Transactions on ML Research](#), 2023
- [23] M Pasquet, ... AT Chieco, **S Dillavou**, JM Hanlan, DJ Durian, E Rio, A Salonen, D Langevin. *Aqueous Foams in Microgravity, Measuring Bubble Sizes* [Comptes Rendus. Mécanique](#), 2023
- [24] SCL Durian[✉], **S Dillavou**, K Markin[✉], A Portales[✉], BOT Maldonado, WTM Irvine, PE Arratia, DJ Durian. *Spatters and Spills: Spreading Dynamics for Partially Wetting Droplets* [Physics of Fluids](#), 2022
- [25] JF Wycoff[✉], **S Dillavou**, M Stern, AJ Liu, DJ Durian. *Desynchronous Learning in a Physics-Driven Learning Network* [Journal of Chemical Physics](#), 2022
- [26] M Stern, **S Dillavou**, MZ Miskin, DJ Durian, AJ Liu. *Physical Learning beyond the Quasistatic Limit* [Physical Review Research](#), 2022
- [27] M Stern, **S Dillavou**, MZ Miskin, DJ Durian, AJ Liu. *Out of Equilibrium Learning Dynamics in Physical Allosteric Resistor Networks* [NeurIPS ML & the Physical Sciences Workshop](#), 2021
- [28] S Zheng, **S Dillavou**, JM Kolinski. *Air Mediates the Impact of a Compliant Hemisphere on a Rigid Smooth Surface* [Soft Matter](#), 2021
- [29] T Pilvelait[✉], **S Dillavou**, SM Rubinstein. *Influences of Microcontact Shape on the State of a Frictional Interface* [Physical Review Research](#), 2020
- [30] JL Silverberg, **S Dillavou**[✉], L Bonassar, I Cohen. *Anatomic Variation of Depth-Dependent Mechanical Properties in Neonatal Bovine Articular Cartilage* [J Orthopaedic Res](#), 2013

Patents

- [P] **S Dillavou**, DJ Durian, AJ Liu, M Stern, MZ Miskin. *Coupled Networks for Physics-Based Machine Learning* [US Patent 12462202 B2](#), 2025

Software Packages Authored

Bellybutton – a Python deep learning package for image segmentation, designed for researchers with no coding required. Download available [here](#). Associated publication: [\[8\]](#)

TEACHING AND MENTORING EXPERIENCE

<u>Research Mentee</u>	<u>Career Stage, University</u>	<u>Coauthored Publications</u>	<u>Year(s)</u>
Jacob F Wycoff	Postgrad	[1][25]	2025–26
Alex Roseman	Undergraduate, Yale U		2024
Juan Mendez	Undergraduate, Williams		2024
Evan Stocker	Undergraduate, Pennsylvania State U		2024
Benjamin D Beyer	Undergraduate, U of Pennsylvania	[7][9][10]	2021–24
Jesse M Hanlan	Ph.D. Student, U of Pennsylvania	[4][5][8][17][23]	2020–24
Josue D Ruiz	Undergraduate, U of Pennsylvania		2022–23
Jacob F Wycoff	Undergraduate, U of Pennsylvania	[1][25]	2021–23
Courtney C Jones	Undergraduate, U Maryland	[17]	2022
Alex Gerra	Undergraduate, Moravian U	[17]	2022
Kwame Markin	Undergraduate, Swarthmore	[24]	2021
Adrian Portales	Undergraduate, U Texas Rio Grande	[24]	2021
Sylvia CL Durian	Undergraduate, U Chicago	[24]	2021
Mary Agajanian	Undergraduate, Harvard U	[21]	2019–21
Tom Pilvelait	Undergraduate, Harvard U	[29]	2018–20
Vincent Stin	Undergraduate, ESPCI, Paris		2019
Aijie Xu	Ph.D. Student, Tsinghua U		2017–18
Evgeni Shirman	Undergraduate, Hebrew U, Jerusalem		2016

Teaching

Substitute Lecturer: Analytical Mechanics, Douglas Durian – 30 Undergrads	U Penn, Spring 2024
TA: Introduction to Fluid Mechanics, Shmuel Rubinstein – 60 Undergrads	Harvard U, Spring 2018
Develop new materials, in-class demos, grade assignments, supervise labs and final projects	
Received Bok Center Certificate of Teaching Excellence	
TA: Introduction to Soft Matter, Shmuel Rubinstein – 20 Grad Students	Harvard U, Fall 2015
Write problem sets, develop new materials, teaching section, grading	

Short Courses / Workshops / Tutoring

Improving Presentation & Discussion Through Improvisation – 15 Grad.	Harvard U, Winter 2019
Intro to Long-Form Improvisation – 15 Graduate Students	Harvard U, Winter 2016
Developed and taught custom curricula for both mini-courses	
Improvisational Theater Workshops	Harvard, Tufts, Yale, Cornell, Deloitte, 2012–Present
Designed & taught for 5–50 participants, 6th grade to grad, business professionals, faculty.	
High school/college math, physics, and SAT tutoring; 100s of hours, 30+ students	2010–Present

Pedagogical Training

Teaching and Communicating Physics	Harvard U, Spring 2015
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PRESENTATIONS AND PRESS

[some titles truncated for space]

Invited Talks

Analog Hardware That Trains Itself	Frontiers of Neuromorphic Computing II, Max Planck, 2026
Emergent Learning in Analog Electronic Networks	(AI) ² Research Talk Series, Princeton U, 2025
Self-Learning Networks in the Lab	Beyond von Neumann Computing Wkshp, Aspen, CO, 2025
M.L. as an Experimental Guide	PREM20: Self-Driving Science: AI & AE, San Juan, PR, 2025
Emergent Learning in Analog Electronic Networks	Physics/Math Special Seminar, NYU, 2025
The Effects of Physicality on Emergent Learning	APS March Meeting, Anaheim, CA, 2025
How to Construct a Learning ‘Material’ (Tutorial)	APS March Meeting, Anaheim, CA, 2025
Emergent Physical Learning	Applied Physics Special Seminar, Cornell U, 2024
Emergent Physical Learning	Tufts Condensed Matter Seminar, 2024
Emergent Machine Learning	Inaugural klogW Future Series Seminar, Virtual, 2024
The Metamaterial that Trains Itself	SIAM: Mathematical Aspects of Mat Sci, Pittsburgh, PA, 2024
Emergent Learning in Electronic Networks	Physics Dept Special Seminar, U Chicago, 2024
Supervised Learning as an Emergent Property	Argonne Nat’l Lab Applied AI Series, Virtual, 2023
Evolution of a Learning Material	9th IDMRCS, Chiba, Japan, 2023
Decentralized, Physics-Driven Learning	SPIE Photonics West, San Francisco, CA, 2023
A Physics-Driven Learning Network	Alternative Computing Group Seminar, NIST, 2023
Demonstration of Decentralized, Physics-Driven Learning	Phys Rev Journal Club, Virtual, 2022
Hijacking Physics to Learn for Us	Weekly Seminar, Google Brain, 2022

Using Physics to Learn without a Processor
 Decentralized Physics-Driven Learning
 Hidden Dynamics of Static Friction
 Hidden Dynamics of Static Friction
 Hidden Dynamics of Static Friction
 Static Friction: Aging and Memory
 Memory in Solid-Solid Interfaces

Selected Press

On Physical & Emergent Learning

[Machines That Learn Like Physicists Think](#)

[How a simple circuit could offer an alternative to energy-intensive GPUs](#)

[A first, physical system to learn nonlinear tasks without a traditional processor](#)

[Training neural networks using physical equations of motion](#)

[How to make the universe think for us](#)

[Simple electrical circuit learns on its own – with no help from a computer](#)

[Programming matter to a computer's job](#)

On the Virtual Frame Technique

[Imaging technique lets ordinary cameras capture high-speed images of crack formation](#)

[How to mod a smartphone camera so it shoots a million frames per second](#)

On Memory in Frictional Interfaces

[Friction Remembers Its Origins](#)

[Friction Remembers Its Past](#)

Contributed Talks

Local Learning Rules in Analog Hardware

Studying Granular Clogging with ML as an Experimental Guide

Emergent Learning Via Sequential Error Mode Reduction

1st Prize Statistical & Nonlinear Physics Postdoctoral Speaker Award

Transistor-Based Self-Learning Networks

A Physics-Driven Self-Learning Transistor Network

Contact Distribution Encodes Frictional Strength

Decentralized Physics-Driven Learning

Building a Physical Learning Network

Memory in Solid-Solid Interfaces

Hidden Dynamics of Static Contact and Static Friction

Extreme Mechanics of Elastomer Impact

Two Solids Make a Glass: Memory in Solid-Solid Interfaces

Elastomer Wear: The NBA's Shoe Problem

Memory in the Frictional Interface

APS March Meeting, Chicago, IL, 2022

Physics Seminar, Bucknell U, 2021

Applied Math Seminar, NYU, 2020

Soft Matter Coffee Hour, Princeton U, 2020

Soft Matter Theory Group (U Pennsylvania), Virtual, 2020

Geomechanics Seminar, Pennsylvania State U, 2018

Mechanical Engineering Seminar, EPFL, 2018

APS Research Highlights, 2026

MIT Tech Review, 2024

Penn Today, 2024

PNAS Physics Commentary, 2024

Quanta Magazine, 2022

Science News, 2022

American Physical Society News, 2021

Phys.org, 2019

MIT Tech Review, 2018

American Physical Society Physics Focus, 2018

Physics Today, 2018

Computing with Physical Systems, 2026

NE Granular Mat, Holy Cross, 2024

APS March Mtg, Minneapolis, MN, 2024

APS March Meeting, Las Vegas, NV, 2023

Coherent Network Comp., Stanford U, 2022

APS March Meeting, Chicago, IL, 2022

Rising Stars in Soft & Biological Matter, U Chicago, 2021

APS March Meeting, Virtual, 2021

APS March Meeting, Boston, MA, 2019

Dynamics Days, Northwestern U, 2019

NORA & BASF Collab. Days, U Mass Amherst, 2019

APS March Mtg, Los Angeles, CA, 2018

NORA & BASF Collab. Days, U Mass Amherst, 2017

Physics Dept Mini-Symposium, Weizmann Inst, 2017

Posters/Rapid Talks

Nonlinear Classification Without a Processor	Ctr for Soft and Living Matter Kickoff, UPenn, 2024
Self-Learning Electronic Networks	Mid-Atlantic Soft Matter Workshop, Georgetown U, 2024
Nonlinear Classification Without a Processor	Computing with Physical Systems, Aspen, CO, 2024
A Physics-Driven Self-Learning Transistor Network	APS March Meeting, Las Vegas, NV, 2023
1st Prize in the APS March Meeting Postdoctoral Poster Competition	
Physical Learning Machines	Cracking the Glass Problem Simons Mtg, New York, NY, 2022
Building a Physical Learning Network	Northeast Complex Fluids Workshop, Virtual, 2021
Tabletop Nucleation	Southern California Earthquake Center Annual Mtg, Palm Springs, CA, 2019
The Hidden Dynamics of Static Friction	Gordon Conf: Soft Matter Phys., New London, NH, 2019
The Virtual Frame Technique	77th New England Complex Fluids, Harvard U, 2018
Memory in the Frictional Interface	73rd New England Complex Fluids, Harvard U, 2018
Memory in the Frictional Interface	Jay (Fineberg) Fest, Hebrew U in Jerusalem, 2017
Beyond Rate and State: Frictional Memory	Inst. for Study of the Continents Conf, Cornell U, 2017
Wear in Basketball Shoes	NORA & BASF Challenges, U Mass Amherst, 2017
Visualizing Frictional Interfaces	69th New England Complex Fluids, Harvard U, 2016
Visualizing Frictional Interfaces	67th New England Complex Fluids, MIT, 2016
Loading History of Frictional Interfaces	Phys + Mech of Soft Complex Mat, Cargese, France, 2016
Loading History of Frictional Interfaces	Gordon Conference: Tribology, Lewiston, ME, 2016
Visualizing Growth of a Multi-contact Interface	65th NE Complex Fluids, Harvard U, 2015
Aging of Multi-Contact Interfaces	Soft Matter: Friction, Rheology, Tribology, U Florida, 2015

PROFESSIONAL SERVICE**Journal Referee**

Science, Nature [Communications, Electronics], Soft Matter, Physical Review [B, E, Applied, Letters], US Geological Survey Internal, J of Geophysical Research – Solid Earth, NeurIPS Workshops

Outreach

Design and teach U Penn Data Driven Discovery Initiative ML Workshop	2023, 2024, 2025
Philly Materials Day (K–12), design, construct and demo trainable elastic material	2024
Design and teach U Penn REU Machine Learning (ML) Workshop	2022, 2023, 2024
DEEPenn STEM (see below), volunteer, mentor, presenter	2023
“Friction: The surprising unsolved science behind earthquakes and tire treads”	Science Café, Wilmington, Delaware, 2023
Planning committee, volunteer, presenter for the first annual DEEPenn STEM: weekend-long STEM PhD prep/info workshop for ~45 URM college students from around the country	2022
MRSEC National Science Slam: Learning Networks on the Radio	2022
Science in the News Writer, Harvard U	2016–17
Splash at Yale Instructor, grades 7–9 and 10–12, Yale U	2016, 2017

Professional Membership

American Physical Society	2016–Present
APS March Meeting session organizer, chair, and sorter	2023, 2024

Miscellaneous

Part of a collaboration developing a 3D Printer-as-Ventilator during COVID-19 outbreak	2020
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